Reasoning test

For admission to Class 11

About the test

- 15 questions in 15 minutes: 5 each of maths, English and science, mixed together.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers

The questions test reasoning, not memory, using ideas taught up to Class 10. Chapters are from the NCERT (CBSE) books a student joining Class 11 in 2027 will have used; other boards teach the same ideas under their own names.

Maths

Class 6 · Mathematics

- *Whole Numbers* – number patterns and series
- *Playing with Numbers* – primes, factors, HCF and LCM
- *Fractions*
- *Mensuration* – perimeter and area
- *Ratio and Proportion*

Class 7 · Mathematics

- *Fractions and Decimals* – fractions
- *Data Handling* – averages (mean)
- *Simple Equations*
- *Comparing Quantities* – percentages, profit and loss, discounts, interest; ratio and proportion
- *Perimeter and Area*

Class 7 · Science

- *Motion and Time* – speed, distance and time

Class 8 · Mathematics

- *Linear Equations in One Variable* – simple equations
- *Squares and Square Roots* – squares and cubes
- *Cubes and Cube Roots* – squares and cubes
- *Comparing Quantities* – percentages, profit and loss, discounts, interest

Class 10 · Mathematics

- *Arithmetic Progressions* – number patterns and series
- *Probability*

Reasoning, not a textbook chapter: Clocks and calendars · Number codes and puzzles · Everyday units · Multi-step word problems · Time and work, pipes and tanks · Problems on ages · Mixtures · Overlapping groups (Venn diagrams)

English

Grammar, vocabulary and reasoning; nothing from the English reader: Vocabulary: meaning in context, opposites, idioms · Grammar: agreement, prepositions, tenses, linking words · Completing and ordering sentences · Analogies, odd one out, letter series and codes · Reading: inference, tone, assumptions, judging an argument · Logical deduction: statements, arrangements, family relations

Science

Class 6 · Science

- *Fun with Magnets* – magnets
- *Light, Shadows and Reflections* – light, shadows and plane mirrors
- *Electricity and Circuits* – electric circuits

Class 7 · Science

- *Heat* – heat and conductors
- *Acids, Bases and Salts* – acids, bases and indicators
- *Physical and Chemical Changes* – physical and chemical changes, rusting
- *Nutrition in Plants* – photosynthesis and plant respiration
- *Respiration in Organisms* – respiration; photosynthesis and plant respiration
- *Motion and Time* – speed
- *Light* – light, shadows and plane mirrors

Class 8 · Science

- *Force and Pressure* – force and pressure, pressure in liquids
- *Sound* – sound, and the speeds of light and sound

Class 9 · Science

- *Matter in Our Surroundings* – evaporation and condensation; change of state
- *Is Matter Around Us Pure?* – elements, compounds, mixtures and solutions
- *Force and Laws of Motion* – inertia, action and reaction
- *Gravitation* – mass and weight; floating, sinking and density
- *Work and Energy* – energy changes

Class 10 · Science

- *Electricity* – electric power and safety
- *Our Environment* – food chains

Reasoning, not a textbook chapter: Reading an experiment and fair tests · Applying a principle to a new situation

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS
   Which of these numbers is divisible by both 3 and 4?

   A 132 **B** 138
   C 142 **D** 150

2. ENGLISH
   The sentries stayed vigilant all night. As used here, "vigilant" most nearly means

   A tired **B** watchful
   C silent **D** brave

3. SCIENCE
   A 1,000 W room heater is switched on for 3 hours. How many units (kilowatt-hours) of electricity does it use?

   Type-in answer: _____

4. MATHS
   Kiran scored 54 marks out of 75 in a test. What percentage did Kiran score?

   Type-in answer: _____

5. ENGLISH
   Choose the correct word: The cadets were proud ___ their achievement.

   A of **B** for
   C with **D** at

6. SCIENCE
   A fuse is fitted in a household circuit so that, if too much current flows, it

   A raises the voltage to appliances **B** melts and breaks the circuit
   C stores the extra electricity **D** makes the bulbs glow brighter

7. MATHS
   A bus leaves at 9:40 am and travels 150 km at a steady 60 km/h. At what time does it arrive?

   A 11:40 am **B** 11:50 am
   C 12:10 pm **D** 12:30 pm

8. ENGLISH
   Which word is the odd one out?

   A Cricket **B** Hockey
   C Football **D** Chess

9. SCIENCE
   Which of these is a compound?

   A water **B** oxygen
   C iron **D** air

10. MATHS

After a 25% increase, a salary is ₹20,000 a month. What was it before the increase?

A ₹15,000
B ₹16,000
C ₹16,500
D ₹17,500

11. ENGLISH

If COLD is written as DPME in a code, how is WARM written in the same code?

A VZQL
B XBSM
C XBSN
D YCTO

12. SCIENCE

A 60 W bulb and a 1,500 W heater are each used for 2 hours. How many times more energy does the heater use?

A 2.5 times
B 12.5 times
C 25 times
D 50 times

13. MATHS

Two fair dice are thrown together. What is the probability that the two numbers add up to 8?

A 1/6
B 1/9
C 7/36
D 5/36

14. ENGLISH

All cadets wake at 5 am. Nobody who wakes at 5 am misses breakfast. Arun misses breakfast. Which statement must be true?

A Arun wakes at 5 am
B Arun is a cadet
C Some cadets miss breakfast
D Arun is not a cadet

15. SCIENCE

A wooden block floats in water with exactly half of its volume under the surface. Water has a density of 1 g/cm³. What is the density of the wood?

A 1 g/cm³
B 2 g/cm³
C 0.25 g/cm³
D 0.5 g/cm³

Answers

1	**A. 132**	132 = 12 × 11, so it is divisible by both. 138 and 150 are divisible by 3 but not by 4; 142 is divisible by neither.
2	**B. watchful**	Vigilant means keeping careful watch for danger, which is exactly what a sentry does. Tired or silent sentries may still be vigilant.
3	**3**	1,000 W is 1 kW, and energy is power × time: 1 kW × 3 h = 3 kWh, which is 3 units. This is a type-in question.
4	**72**	54 ÷ 75 = 0.72, which is 72%. Another way: 75 marks is 100%, so each mark is 4/3 %, and 54 × 4/3 = 72. This is a type-in question.
5	**A. of**	"Proud" is followed by "of": proud of something. The other prepositions are not used with proud in this sense.
6	**B. melts and breaks the circuit**	A fuse wire melts when the current is too large, which breaks the circuit and stops the flow before wires overheat or catch fire.
7	**C. 12:10 pm**	The journey takes 150 ÷ 60 = 2.5 hours, which is 2 hours 30 minutes. 9:40 am plus 2 hours 30 minutes is 12:10 pm.
8	**D. Chess**	Cricket, hockey and football are outdoor team games played with a ball. Chess is a board game for two players.
9	**A. water**	Water is hydrogen and oxygen chemically joined, so it is a compound. Oxygen and iron are elements, and air is a mixture.
10	**B. ₹16,000**	The new salary is 125% of the old one, so the old salary is 20,000 ÷ 1.25 = ₹16,000. Taking 25% off ₹20,000 gives the wrong ₹15,000.
11	**C. XBSN**	Each letter moves one step forward: C→D, O→P, L→M, D→E. So W→X, A→B, R→S, M→N, giving XBSN.
12	**C. 25 times**	Both run for the same time, so energy goes up with power alone: 1,500 ÷ 60 = 25. The 2 hours cancels out.
13	**D. 5/36**	There are 36 equally likely outcomes. A total of 8 comes from (2,6), (3,5), (4,4), (5,3) and (6,2), which is 5 outcomes: 5/36.
14	**D. Arun is not a cadet**	Every cadet wakes at 5 am, and everyone who does has breakfast. Arun does not, so Arun cannot wake at 5 am and so cannot be a cadet.
15	**D. 0.5 g/cm^3**	A floating block pushes aside its own weight of water. It does so with half its volume, so it weighs as much as half its volume of water: 0.5 g/cm^3.